

Chapter 48: Multifetal Pregnancy — Revision Notes

Williams Obstetrics 26e, pp. 2182–2254 of the PDF. High-yield numbers are in **bold**. Both tables are transcribed from the PDF images (they are missing from the extracted chapter text). 13 of the 17 figures are included.

Big picture: Chorionicity, not zygosity, drives most twin-specific risk. Monochorionic twins share placental vessels, which causes TTTS, TAPS, TRAP and co-twin injury after a demise. Preterm birth is the dominant problem for all multiples, and almost nothing prevents it.

- US twin birth rate **3.2%** (2019). Triplets-or-more fell **55%** from their 1998 peak (0.2% of births).
- Multiples were **3% of live births but 15% of infant deaths**. Infant mortality is **>4x for twins** and **12x for triplets**.
- Mother: preeclampsia, PPH and maternal death **2x singletons**. Peripartum hysterectomy **3x (twins), 24x (triplets/quads)**. ↑ previa and accreta spectrum.

Table 48-1. Selected Outcomes in Singleton and Twin Pregnancies Delivered at Parkland Hospital from 1988 through 2021

Outcome	Singletons (No.)	Twins (No.)
Pregnancies	456,120	5366
Births ^{a b}	456,120	10,732
Stillbirths	2532 (5.6)	258 (24.0)
Neonatal deaths	918 (2.1)	186 (18.2)
Perinatal deaths	3453 (7.6)	444 (41.4)
Very low birthweight (<1500 g)	4450 (10.0)	1109 (108.8)

^aBirth data are number (per 1000). ^bDenominator for neonatal deaths and very low birthweight is liveborn infants.

1. Mechanisms of Multifetal Gestations

Dizygotic vs monozygotic

- Describe by **zygosity, amnionicity and chorionicity** (number of zygotes, amnions, chorions).
- **Dizygotic (fraternal):** two ova fertilized in one cycle. Genetically just siblings. **Opposite sex = almost always dizygotic**.
 - Rare exceptions: postzygotic Y loss in one 46,XY monozygotic twin → 45,X Turner female twin.
- **Monozygotic (identical):** one zygote divides. Still not truly identical: unequal protoplasm, postzygotic mutations, variable expression, **skewed lyonization** in females.
- **Sesquizygosity:** rare dispermic fertilization that yields monozygotic twins; may be sex-discordant.

Monozygotic twinning: timing of division

- Association with **ART/IVF:** monozygotic twinning **2x more common with blastocyst transfer** (culture media effect?).

Division after fertilization	Result
<72 h (0–4 days)	Dichorionic diamniotic (placentas separate or fused)
4–8 days	Monochorionic diamniotic
8–12 days	Monochorionic monoamniotic
Later (>13 days)	Conjoined twins

- Rarely, **monochorionic twins are dizygotic**: ~80% after ART, 90% with **chimerism**.

Memory hook: "3-8-12": split by day **3** = two of everything; by day **8** = shared chorion; by day **12** = shared amnion; later = conjoined.

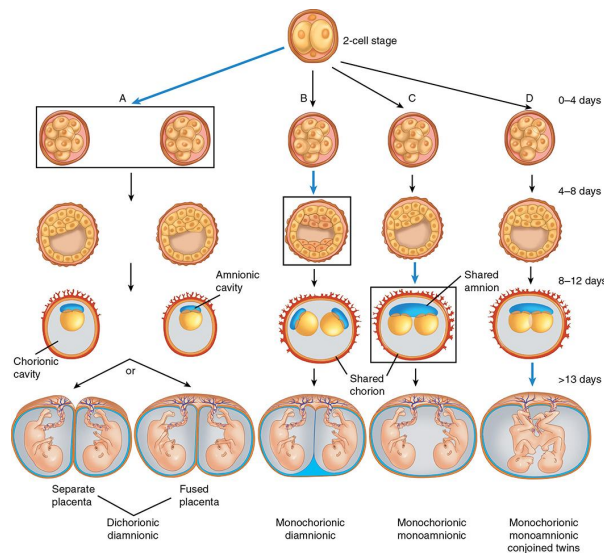


Figure 48-1. Timing of monozygotic division determines chorionicity and amnionicity: DCDA → MCDA → MCMA → conjoined.

Factors affecting twinning

- **Monozygotic rate is constant worldwide: ~1 in 250 births**, independent of demographics except ART.
- **Dizygotic twinning** ↑ with: **infertility treatment, maternal age, race, heredity, maternal size/nutrition**.
- **ART**: clomiphene or FSH + hCG → multiple ovulation. More embryos transferred → more multiples.
 - ART = **1.9%** of US newborns but **14.7%** of multifetal neonates (2017).
 - ASRM: women **<35 y** → **single-embryo transfer**, whatever the embryo stage.
- **Age**: rising **FSH** with age → greater ovarian stimulation. Paternal age effect is small.
- **Race (US 2019)**: black **4.1%**, white **3.3%**, Hispanic **2.5%**. Rural Nigeria **1 in 20** births.
- **Heredity (maternal side dominates)**: a woman who is herself a DZ twin → **1 in 58**; husband a DZ twin → 1 in 116.
- **Size/nutrition**: tall, heavy women **25–30%** higher rate. WWII undernourishment → fall in DZ twinning (nutrition > body size).

Superfecundation and superfetation

- **Superfecundation**: two ova fertilized in the same cycle but not at the same coitus, possibly by different men (**heteropaternality**).
- **Superfetation**: ≥1 menstrual cycle between fertilizations. Not known spontaneously; likely due to ART.
- **Pseudosuperfetation**: markedly unequal growth of twins of the same age.

2. Diagnosis of Multifetal Gestation

Clinical

- Fundal height at 20–30 wk averages **5 cm greater** than singletons.
- Palpating **two fetal heads** strongly supports it; Doppler may find two distinct heart rates.
- Clinical criteria alone are unreliable: in the **RADIUS** trial, unscanned twins were undiagnosed until **26 wk in 37%** and until **delivery admission in 13%**.

Sonography

- Aim to define **fetal number, gestational age, chorionicity, amnionicity**. See each head in two perpendicular planes; ideally both heads or abdomens in one image.
- **Accuracy for chorionicity: 98% in the 1st trimester**; wrong in up to **10%** in the 2nd; **~95%** correct before 24 wk. Membrane assessment establishes chorionicity in **>99%** in the 1st trimester.
- **Early 1st trimester**: number of **chorions = number of gestational sacs**. Thin amnion may be invisible **before 8 wk** → **number of yolk sacs ≈ number of amnions** (not always accurate). Cord entanglement = monoamnionic.
- **10–14 wk, four features**:
 1. **Number of placental masses** (two = dichorionic; one fused mass doesn't exclude it)
 2. **Intervening membrane** present?
 3. **Membrane thickness: ≥ 2 mm = dichorionic** (4 layers: 2 amnions + 2 chorions); **< 2 mm = monochorionic**
 4. **Fetal sex**: discordant = dichorionic (and dizygotic)

Sign	What it is	Means
Twin-peak (lambda, delta) sign	Triangular projection of placenta between the layers of the dividing membrane	Dichorionic
T sign	Thin membrane meets placenta at a right angle, no tissue between	Monochorionic diamnionic
No dividing membrane	—	Monochorionic monoamnionic

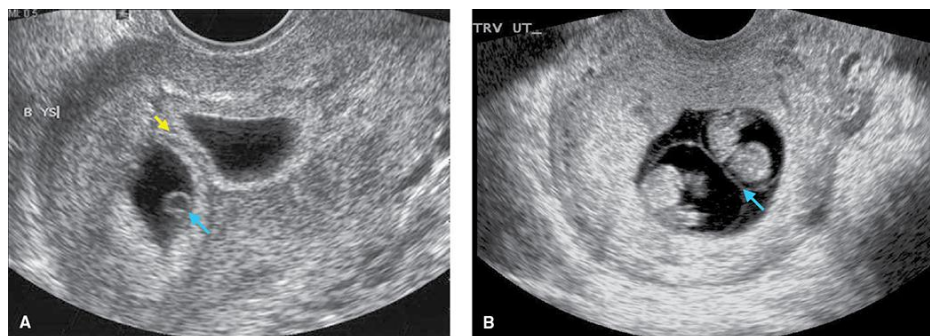


Figure 48-2. First-trimester twins: (A) DCDA at 6 wk with a thick dividing chorion; (B) MCDA at 8 wk with a thin dividing membrane.

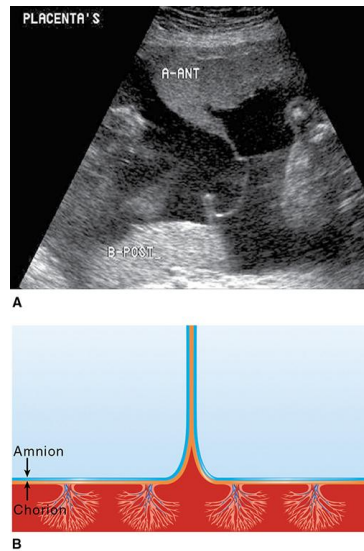


Figure 48-3. Twin-peak (lambda) sign: placental tissue insinuates between the amniochorion layers → dichorionic.

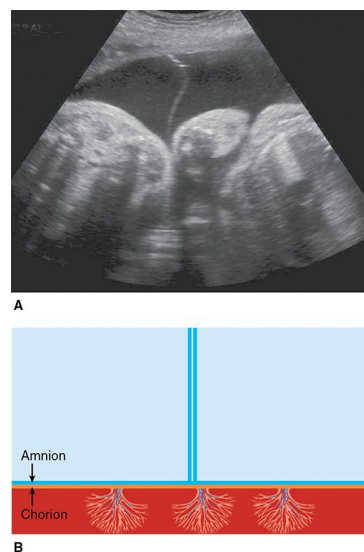


Figure 48-4. "T" sign: two apposed amnions meet the placenta with no intervening chorion → monochorionic diamnionic.

Other aids

- **MRI** helps delineate monochorionic complications, including conjoined twins.
- Abdominal radiography: limited use (fetal movement → errors).
- **No biochemical test** reliably diagnoses multiples: β -hCG and MSAFP are higher but overlap with singletons.

Placental examination

- One common amniotic sac, or apposed amnions **without chorion** between → **monozygotic**.
- Amnions separated by **chorion** → DZ or MZ, but **DZ is more common**.
- Different sex or blood types = dizygotic. Same sex/type does **not** confirm monozygosity. Postnatal zygosity DNA testing is available.

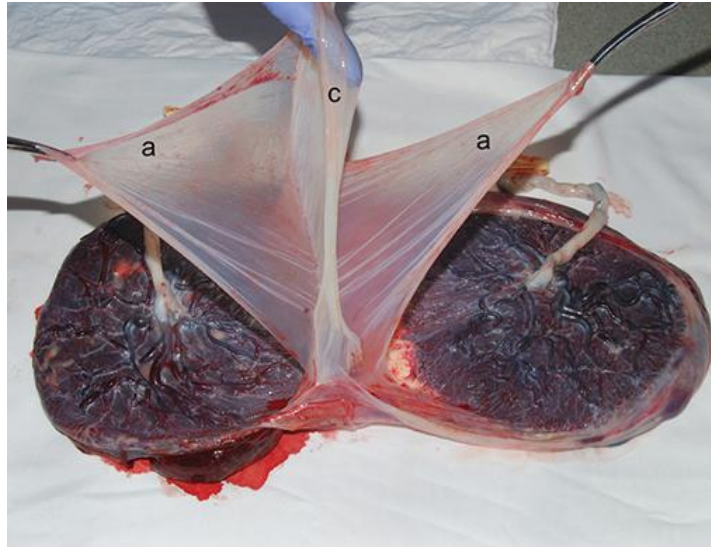


Figure 48-5. DCDA placenta: the dividing membrane has chorion (c) between two amnions (a).

3. Maternal Physiological Adaptations

- $\uparrow$ β -hCG $\rightarrow$ more **nausea and vomiting** in the 1st trimester.
- **Blood volume** $\uparrow$ **50–60%** (vs 40–50% singleton). Offsets vaginal-delivery blood loss, which is **2x** that of a singleton.
- Red cell mass rises proportionately less + higher **iron and folate** needs $\rightarrow$ **anemia**.
- **BP**: diastolic is **lower from 8 wk**, then rises more at term: a rise of **≥ 15 mm Hg in 95%** of twins vs 54% of singletons.
- **Cardiac output** $\uparrow$ **a further ~20%** above singletons, mainly from **stroke volume** (not heart rate). SVR lower throughout. Progressive **diastolic dysfunction** that normalizes after delivery.
- Uterus + nonfetal contents may reach **≥ 10 L** and weigh **>20 lb**. Hydramnios (especially monozygotic) $\rightarrow$ compression of viscera and lungs; rarely obstructive uropathy with renal impairment.
 - **Therapeutic amniocentesis** can relieve symptoms, but fluid reaccumulates quickly.

4. Pregnancy Complications

Spontaneous abortion and vanishing twin

- **Monochorionic** twins have higher early loss than dichorionic.
- **1 in 8** pregnancies start multifetal, but only **1 in 80** births are multifetal.
- One twin "**vanishes**" before the 2nd trimester in **10–40%** of twin pregnancies (more after ART).
- Spontaneous loss of ≥ 1 embryo before 12 wk: **36% of twins, 53% of triplets, 65% of quadruplets**.
- **Effect on screening**:
 - Loss **before 9 wk**: 1st-trimester markers unaffected. **After 9 wk**: markers higher and less precise.
 - **PAPP-A** $\uparrow$ (1st trimester); **MSAFP and inhibin A** $\uparrow$ (2nd trimester).
 - **cfDNA**: a vanishing twin caused **15%** of false-positive results. Identify the reduced sac to avoid confusion.

Congenital malformations

- **406 vs 238 per 10,000** (twins vs singletons). **Monochorionic** $\approx$ **2x** dichorionic.
- Congenital heart disease **73% higher** in twins; higher concordance in monozygotic.
- European rates rose 2.2% $\rightarrow$ 3.3% (1987–2007) with more DZ (ART) twins. After adjusting for maternal age/subfertility, ART itself doesn't clearly $\uparrow$ anomalies.

Low birthweight

- Twin weights track singletons until **28–30 wk**, then lag; clearly diverge from **35–36 wk**. → Use **twin-specific growth curves** (stratified by chorionicity).
- Parkland surveillance: **dichorionic** sonography **every 4 wk from 16–20 wk**; **monochorionic every 2 wk** (TTTS).

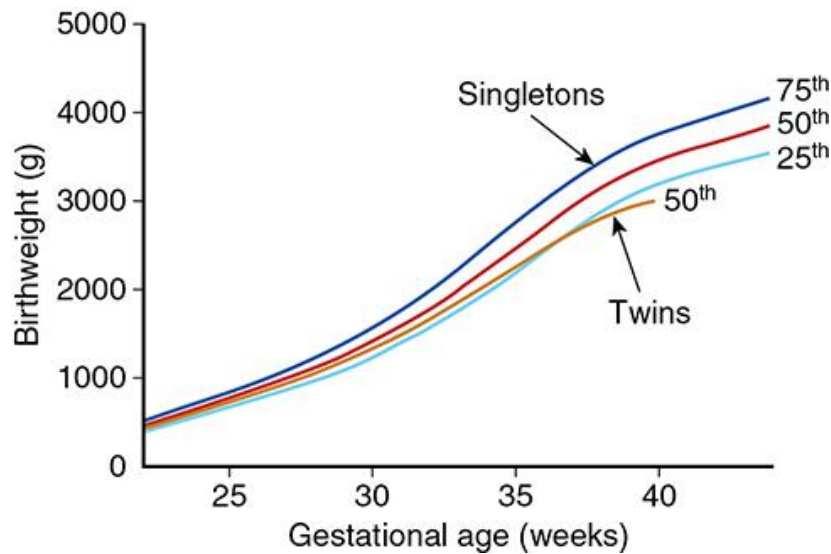


Figure 48-6. Twin 50th-percentile birthweight falls below the singleton curve after ~30 wk and diverges clearly from 35–36 wk.

Hypertension

- Risk rises with fetal number: **14%** of twin pregnancies develop a hypertensive disorder. No zygosity effect. Twins + GDM → more preeclampsia.
- Twins have ↑ **sFlt-1 and soluble endoglin**, ↓ **PIGF** even without hypertension; with preeclampsia, sFlt-1 and sFlt-1/PIGF rise further.
- **Low-dose aspirin** recommended (see Prenatal Care).

Long-term development

- Academic performance similar to singletons, including after ART.
- **Cerebral palsy: 7 vs 1.6 per 1000** live births (multiples vs singletons); 2× even among normal-birthweight neonates. Mainly from **prematurity**; also FGR, anomalies, TTTS, co-twin demise.

Table 48-2. Incidence of Some Complications Related to Twin Characteristics

Rates of twin-specific complications, in percent:

Type of Twinning	Twins	Fetal-Growth Restriction	Preterm Delivery ^a	Placental Vascular Anastomosis	Perinatal Mortality
Dizygotic	80	25	40	0	10–12
Monozygotic	20	40	50	—	15–18
Diamnionic/dichorionic	6–7	30	40	0	18–20
Diamnionic/monochorionic	13–14	50	60	100	30–40
Monoamnionic/monochorionic	<1	40	60–70	80–90	58–60
Conjoined	0.002 to 0.008	—	70–80	100	70–90

^aDelivery before 37 weeks. The three indented rows are subtypes of monozygotic twins.

5. Unique Fetal Complications

Most twinning-specific complications occur in **monozygotic (monochorionic)** twins.

Monoamniotic twins

- **<1% of twins, 4% of monochorionic** pairs.
- Of fetuses alive before 16 wk, **less than half** survive to the neonatal period (anomalies, miscarriage).
- Deaths: preterm birth, TTTS, **cord entanglement**, discordance, anomalies.
- **Anomalies 18–28%** (concordant in only ~1/4) → **fetal echocardiography**.
- Down syndrome risk per fetus $\leq$ age-matched singletons.
- **TTTS is less common** than in MCDA: near-universal **artery-to-artery anastomoses are protective**. Still screen.
- **Cord entanglement** is frequent; color Doppler shows it, but death from it is **unpredictable**.
- **Inpatient vs outpatient surveillance:**
 - Heyborne: **0 stillbirths** in 43 admitted at 26–27 wk vs **13 deaths** in 44 outpatients.
 - MONOMONO: fetal death **3.3% inpatient** (admitted 24–29 wk) vs **10.8% outpatient**. **No deaths after 32 wk.**
- **Management (US):** admit at **24–28 wk** for **1 h daily FHR monitoring** (NST/BPP), betamethasone, **cesarean at 32⁰/₇–34⁰/₇ wk**.

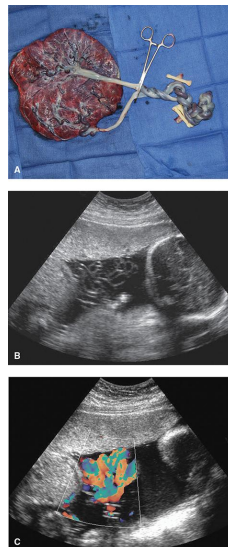


Figure 48-7. MCMA cord entanglement: knotted cords at cesarean (A), entwined cords on sonography (B), accentuated by color Doppler (C).

Aberrant twinning

Spectrum from incomplete splitting (or secondary fusion). Asymmetrical forms: **external parasitic twin, fetus-in-fetu, TRAP sequence**.

Conjoined twins

- **1.5 per 100,000** births. **Thoracopagus is most common**.
- Named by joined region: ventral (rostral: omphalo-, thoraco-, cephalopagus; caudal: ischiopagus; lateral: parapagus) and dorsal (craniopagus, rachipagus, pygopagus).
- **1st-trimester sonographic clues:**
 - Fetal poles closely associated and **don't change relative position**
 - **>3 vessels** in the cord
 - Fewer limbs than expected
 - Spine hyperflexion, bifid fetal pole, ↑ nuchal thickness
- 3D US, color Doppler and MRI clarify shared organs.

- Separation succeeds if essential organs are not shared. **Viable conjoined twins → cesarean.** For termination, vaginal delivery is possible (union is pliable) but dystocia and uterine/cervical trauma are common.

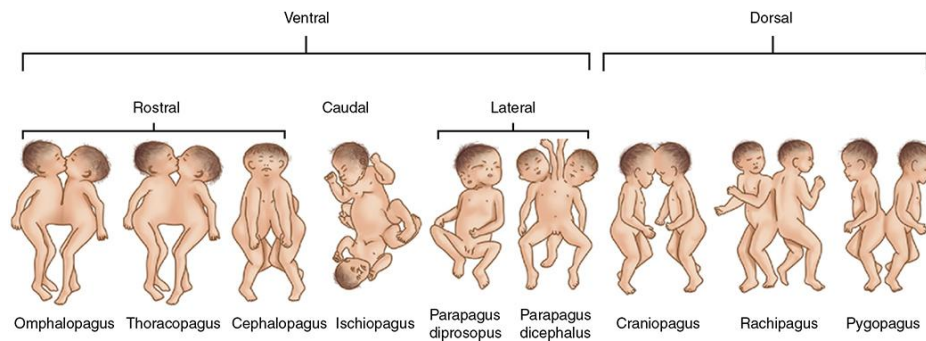


Figure 48-8. Types of conjoined twins: ventral (rostral, caudal, lateral) and dorsal unions. Thoracopagus is most common.

External parasitic twin

- Grossly defective fetus or parts (supernumerary limbs ± viscera) attached to a normal twin; **functional heart and brain absent. 4%** of conjoined twins.

Fetus-in-fetu

- Embryo enfolded within its co-twin, mainly **intraabdominal**; arrests in the 1st trimester. **Vertebral/axial bones present; heart and brain absent.** Thought to be MCDA, supplied by large parasitic vessels.

Monochorionic twins and vascular anastomoses

- **Virtually all monochorionic placentas** have anastomoses (median 8); they are essentially unique to monochorionic twins.
- **Superficial** (chorionic surface): **artery-artery (AA), vein-vein (VV), artery-vein.**
- **Deep AV anastomoses** run through a shared cotyledon → "**third circulation**" (common villous district), found in ~1/2.
- Danger depends on hemodynamic balance. Imbalance → **TTTS, TAPS, TRAP.**

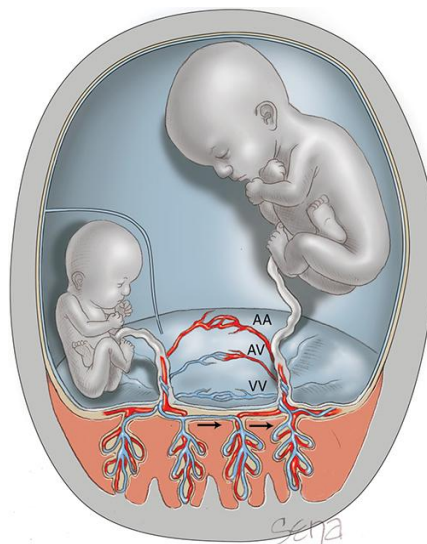


Figure 48-11. AA, AV and VV anastomoses. A deep AV "third circulation" shunts blood from the small "stuck" donor to the recipient.

Twin-twin transfusion syndrome (TTTS)

- **10–15% of monochorionic twins.**
- **Mechanism:** unidirectional flow through **deep AV anastomoses** (donor artery → shared cotyledon → recipient vein), uncompensated by superficial AA anastomoses.

Donor	Recipient
Hypovolemic, oliguric → oligohydramnios ("stuck twin")	Hypervolemic, polyuric → hydramnios
Anemic, pale, smaller, growth-restricted	Polycythemic , larger, volume excess
Contractures, pulmonary hypoplasia	Heart failure / hydrops , hyperviscosity, hyperbilirubinemia/kernicterus, PROM

- Presents in **midpregnancy**. Also called "**poly-oli**" (polyhydramnios–oligohydramnios syndrome).

Fetal brain damage

- CP, microcephaly, porencephaly, multicystic encephalomalacia (ischemic necrosis). Cerebral abnormalities in **8%** of liveborn TTTS fetuses.
- **If one twin dies**: blood shifts acutely from the survivor into the low-resistance vessels of the dead twin → **hypovolemia and ischemic brain injury** in the survivor. This happens at the moment of death, so **immediate delivery doesn't help** without another indication.

Diagnosis (SMFM): two criteria

1. **Monochorionic** pregnancy
 2. **Hydramnios: largest vertical pocket >8 cm** in one sac **and oligohydramnios: <2 cm** in the other
- Growth discordance/FGR is **not** a diagnostic criterion.
 - **Surveillance: sonography every 2 wk from ~16 wk** (intervals >2 wk → higher stage at diagnosis).

Quintero staging

Stage	Criteria
I	Discordant fluid (>8 cm / <2 cm); donor bladder still visible
II	Donor bladder not visible
III	Abnormal Doppler : umbilical artery, ductus venosus or umbilical vein
IV	Ascites or hydrops in either twin
V	Demise of either fetus

- Recipient cardiac function also declines (not in Quintero); many centers add fetal echo (e.g., myocardial performance index).



Figure 48-12. TTTS: stage I "stuck" donor wrapped by its membrane (A); recipient pocket >10 cm (B); stage II donor with empty bladder outlined by color Doppler (C).

Management and prognosis

- Stage I: some stay stable or regress, but **60% progress**. **Stage ≥III untreated: perinatal loss 70–100%**.
- **Eurofetus trial** (severe TTTS <26 wk): survival of ≥1 twin to 6 months **76% with laser vs 51% with serial amnioreduction**. No extra survival or neurological benefit at 6 years.
- **Laser ablation of anastomoses = preferred for stages II–IV**. Stage I: laser or expectant care (controversial).
- **Solomon technique** (ablate the whole vascular equator) → less TTTS recurrence and less **post-laser TAPS**.
- After laser: **weekly** US + Doppler (growth, fluid, placental function, anemia).

- **Selective reduction** (considered if severe before 20 wk): must occlude the **cord/umbilical vein** (RFA, fetoscopic ligation, laser/bipolar coagulation) because injected agents reach the co-twin. Termination of the whole pregnancy is another option.

Twin anemia–polycythemia sequence (TAPS)

- Chronic fetofetal transfusion with large **Hb difference but NO fluid discordance**.
- **Spontaneous: 1–6%** of monochorionic twins, any gestational age.
- **Iatrogenic: up to 13% after laser**, usually within **5 wk**. The former **recipient becomes anemic** and the former donor polycythemic.
- **Diagnosis: MCA-PSV >1.5 MoM in the donor and <1.0 MoM in the recipient**.
- Staging exists (higher stage → higher mortality). Screening for spontaneous TAPS is controversial.
- Options: expectant, delivery, laser, intrauterine transfusion, selective feticide, termination. Postnatal: transfusion for the donor, partial exchange for the recipient.

Twin reversed-arterial-perfusion (TRAP) sequence (acardiac twinning)

- **1 in 35,000** births; monochorionic.
- Normal **pump (donor) twin** + **acardiac recipient** perfused via a large **AA (± VV) anastomosis** with **reversed, deoxygenated** arterial blood.
- Blood enters via the recipient's umbilical arteries → **iliac vessels** → only the **lower body is perfused**; upper body fails to develop.
 - **Acardius acephalus** (no head); **acardius myelacephalus** (partial head, identifiable limbs); **acardius amorphous** (no recognizable structure).
- Pump twin → **cardiomegaly, high-output heart failure**. Historic pump-twin mortality **>50%**; risk ∝ **size of the acardiac twin**.
- **Radiofrequency ablation (RFA) of the cord** is preferred: NAFTNet median delivery **37 wk, 80%** donor survival.

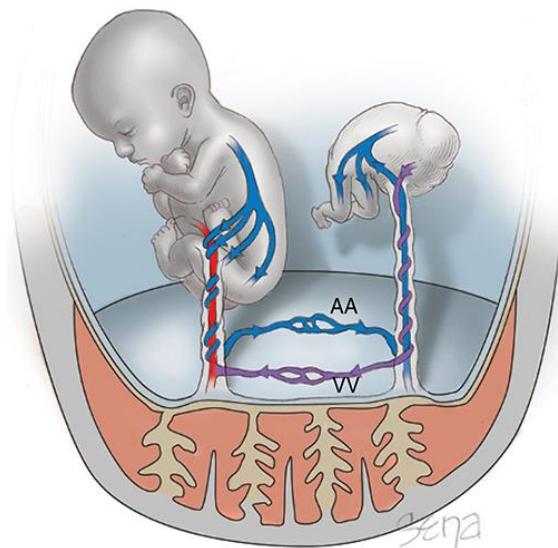


Figure 48-13. TRAP: AA and VV anastomoses let the pump twin perfuse the acardiac twin's lower body with reversed, deoxygenated blood.

Complete mole with coexisting normal fetus

- One normal fetus + a **complete mole** co-twin. Differentiate from **partial mole** (singleton **triploid** fetus with molar placenta).
- Sonography: normal twin + large placenta with **multiple small anechoic cysts**.
- Continuing is increasingly accepted: **live birth ~50%**. **GTN risk is similar whether terminated or not**.
- Risks of continuing: bleeding, hyperemesis, **thyrotoxicosis, early-onset preeclampsia** → preterm birth. **Postpartum GTN surveillance is essential**.

6. Discordant Growth

- ~**15%** of twins. Mortality rises with the weight difference; worse if **early** and **monochorionic**. Discordance at ≤ 20 wk $\rightarrow$ **8–15%** of the smaller fetuses die.

Pathogenesis

Monochorionic	Dichorionic
Unequally shared placenta (main cause)	Different genetic growth potential (esp. opposite sex)
TTTS: anastomotic perfusion imbalance	Suboptimal implantation site for one placenta
Discordant structural anomalies	Cord abnormalities: velamentous/marginal insertion, vasa previa
	Crowding (severe discordance 2x in triplets); placental histological lesions

Diagnosis

- CRL differences **don't** predict birthweight discordance $\rightarrow$ surveillance starts after the 1st trimester.
- **% discordance = (larger EFW – smaller EFW) / larger EFW.**
- **ACOG: discordance = EFW difference >20%.**

Management

- Discordance between two **appropriately grown** twins may not add risk.
- **>25–30% discordance** best predicts adverse outcome. Parkland (1370 pairs): RDS, IVH, seizures, PVL, sepsis and NEC rose with discordance, sharply **>25%**.
 - **Relative risk of fetal death: 5.6 if >30%; 18.9 if >40%.**
- NST/BPP; umbilical artery Doppler of the smaller **monochorionic** twin. Deliver at advanced gestations.

Selective fetal-growth restriction (sFGR)

- Usually late 2nd/early 3rd trimester. Defined by **AC difference >20 mm** or **discordance >20%**.
- Once diagnosed: **weekly** fetal testing, fluid and umbilical artery Doppler. Growth scans **every 3 wk.**

sFGR type (Gratacós)	Umbilical artery end-diastolic flow (smaller twin)	Course
I	Positive	Smaller discordance, relatively benign
II	Persistently absent	High risk of deterioration and demise
III	Intermittently absent/reversed	Large AA anastomoses $\rightarrow$ lower risk than type II

- Parkland: **daily inpatient surveillance** if one twin is growth-restricted or discordance **>25%** with a growth-restricted twin.

7. Fetal Demise

- Causes: anomalies and chorionicity (sFGR, TTTS, TAPS in MC pairs; cord entanglement in MCMA).
- **Early loss** = vanishing twin: survivor's risk is **not** increased; no extra surveillance needed.
- Later: **fetus compressus** (compressed) or **fetus papyraceus** (flattened and desiccated).

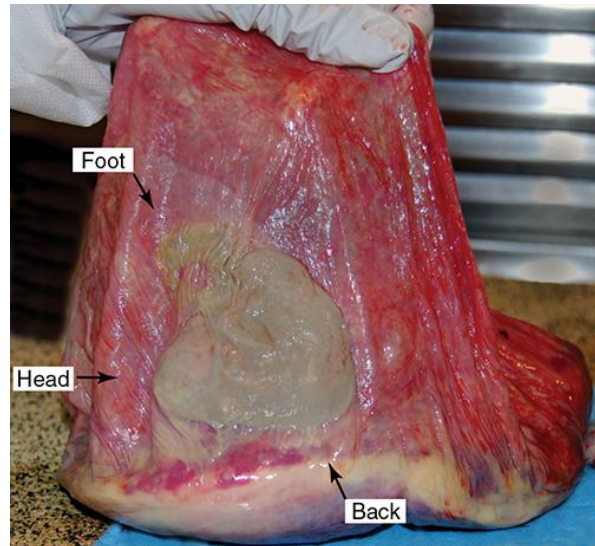


Figure 48-16. Fetus papyraceus: a tan, flattened ovoid mass on the membranes after demise at ~17 wk; the co-twin delivered at 40 wk.

Single intrauterine fetal demise (sIUFD) after 20 wk

- Twin stillbirth rate **1.4%** vs singleton **0.6%**.
- Risks to the survivor: **death, preterm birth, neurological injury** (acute hypovolemia through anastomoses at the moment of death; not preventable).
- **Co-twin death: MCDA 16× more likely** than DC twins. Risk falls after **26–28 wk**.
- Preterm birth ↑, similar in MC and DC.
- **Neurodevelopmental morbidity (sIUFD <34 wk): 29% in monochorionic, ~3× dichorionic.** After 34 wk: similar.
- **Management:**
 - MC demise after the 1st trimester but **before viability** → termination can be considered.
 - After viability: prolong gestation, prioritize maternal/co-twin safety, consider corticosteroids. **Immediate delivery isn't beneficial** without another indication.
 - **sIUFD ≥34 wk → delivery is reasonable.**
- **Maternal coagulopathy** is rare (the survivor is usually delivered within weeks).

8. Prenatal Care

- Parkland: visits **every 2 wk from 22 wk**, with **digital cervical exam** each visit.
- **Weight gain (IOM, normal BMI): 37–54 lb.**
- **Calories: 40–45 kcal/kg/d;** diet **20% protein, 40% carbohydrate, 40% fat**, as 3 meals + 3 snacks.
- Supplement calcium, magnesium, zinc, vitamins C, D, E.
- **Low-dose aspirin 81 mg/d from 12–28 wk until delivery** (multifetal = high risk for preeclampsia).
- **Aneuploidy screening:** combined/serum screening with multifetal thresholds; **cfDNA acceptable** (vanishing twin caveat).
- Midpregnancy anatomy scan; **fetal echo for monochorionic** twins.
- **Amniotic fluid: deepest vertical pocket per sac. <2 cm = oligohydramnios, >8 cm = hydramnios.**
- Stillbirth: monochorionic **2–3×** dichorionic, including beyond 34 wk.
- **Weekly antenatal testing (NST/BPP):** from **36 wk for uncomplicated DC** and **32 wk for uncomplicated MC** twins. High false-positive rates → counsel first (iatrogenic preterm birth).

9. Preterm Birth

- US preterm birth rate: **twins 60%, triplets 98%** (2019).
- Causes in twins: **~60% indicated, 1/3 spontaneous labor, 10% PPROM**.
- Higher with monochorionic; also prior preterm birth, adolescence, nulliparity, obesity, diabetes.

Prediction

- Cervical length at 22–24 wk vs delivery <32 wk: **10 mm → 66%; 20 mm → 24%; 40 mm → 1%**.
- **CL <20 mm** best predicts birth <34 wk: specificity **97%**, LR+ **9.0**.
- A closed internal os on digital exam predicted as well as normal CL + negative fetal fibronectin.
- **But screening doesn't improve outcomes:** SMFM recommends **against routine CL screening** in multiples; **fFN not recommended**.

Prevention: nothing works reliably

Strategy	Verdict
Bed rest / hospitalization	No benefit (almost half of outpatients were admitted anyway for indications)
Prophylactic oral β -mimetics	Ineffective; FDA warning against oral terbutaline
17-OHPC	Not effective , even with CL <36 mm or <25 mm
Vaginal progesterone	Ineffective in general twin population; short-cervix data conflicting. Not used at Parkland
Prophylactic cerclage	No benefit (with or without short cervix)
Physical exam–indicated (rescue) cerclage	May benefit: dilated cervix <24 wk + antibiotics + indomethacin → less preterm birth and perinatal death
Arabin pessary	Mixed trials; not recommended by ACOG

Treatment

- **Tocolysis does not improve neonatal outcomes** and is riskier: hypervolemia → **pulmonary edema**.
 - β -mimetics: cardiovascular complications **43% vs 4%** (twins vs singletons). Nifedipine: tachycardia **19% vs 9%**.
- **Antenatal corticosteroids:** less well studied; mixed results. **Use as for singletons.**

PPROM

- Proportion of preterm births with PPROM: singletons **13%**; twins **17%**; triplets and quads **20%**; higher-order **100%**.
- Expectant management is comparable to singletons, but **latency is shorter:** median **4 days (twins) vs 7 days** after 24 wk; shorter still after 30 wk.

Delayed delivery of the second twin

- Maternal complications in **~40%** (infection, PPH, abruption).
- Median latency **6 days** (range 6–107). Retained fetuses survive better than the first-born.
 - First-born <25 wk: **0% survived vs 50%** of co-twins. >25 wk: **65% vs 95%**.
- Cerclage, antibiotics, tocolysis didn't improve outcome. Most likely to benefit at **22–24 wk**. Good candidates are rare.

10. Labor and Delivery

Delivery timing (ACOG)

Twin type (uncomplicated)	Deliver at
Dichorionic diamniotic	38 ⁰ / ₇ –38 ⁶ / ₇ wk
Monochorionic diamniotic	34 ⁰ / ₇ –37 ⁶ / ₇ wk (Parkland: not before 37 wk without another indication)
Monochorionic monoamniotic	32 ⁰ / ₇ –34 ⁰ / ₇ wk (cesarean)

- Stillbirth peaks: DCDA **10.6/1000 at 38⁶/₇ wk**; MCDA **9.6/1000 at 37⁶/₇ wk**.
- Lung maturity is usually synchronous and **several weeks earlier** than in singletons. In sFGR the smaller twin has less RDS but more BPD.

Preparations

Higher rates of: uterine dysfunction, malpresentation, **cord prolapse**, previa, abruption, emergency operative delivery, **atonic PPH**. **Second twins** have worse composite outcomes regardless of route.

1. Trained attendant throughout; **continuous electronic monitoring** (internal for the presenting twin once possible)
2. IV line for rapid infusion; LR or dextrose at **60–125 mL/h**
3. **Blood available**
4. Obstetrician skilled in **intrauterine manipulation**
5. **Sonography machine** in the room
6. **Anesthesia** immediately available
7. **One neonatal resuscitator per fetus**
8. Adequate space and equipment

Analgesia and anesthesia

- **Labor epidural is ideal**: can be extended quickly for internal podalic version or cesarean.
- Uterine relaxation for manipulation: **halogenated inhalation agent** (general anesthesia), or IV/sublingual **nitroglycerin** or IV **terbutaline**.

Presentation

- Commonest: **cephalic–cephalic, cephalic–breech, cephalic–transverse**. First twin cephalic in **71%** at admission (Parkland).
- The second twin is unstable: compound, footling breech, face/brow, cord prolapse (more with small fetuses, hydramnios, high parity).

Delivery route

Presentation	Approach
Cephalic–cephalic	Vaginal birth near term; planned cesarean doesn't help; trial of labor succeeds in ~80%
Cephalic–noncephalic	Controversial. Vaginal delivery reasonable if each >1500 g and >32 wk with skilled provider (breech extraction ± internal podalic version); or cesarean for both
Breech first twin	Usually cesarean (Parkland practice); vaginal delivery is supported if >32 wk and >1500 g

- **Cephalic–noncephalic evidence**:
 - Schmitz (5915): 75% planned vaginal, **80%** successful. Planned cesarean <37 wk had *higher* perinatal morbidity/mortality (**5.2% vs 3.0%**).
 - **Twin Birth Study**: planned vaginal = planned cesarean outcomes. Cephalic vs noncephalic second twins did equally well when delivered vaginally.

- For the second twin, **internal podalic version** beats intrapartum external cephalic version (ECV → more cesareans and fetal distress).
- **Worst scenario:** vaginal first twin + **cesarean for the second** (cord prolapse, abruption, contracting cervix, nonreassuring FHR).
- **Breech first twin problems:** **head entrapment** above an incompletely dilated cervix (preterm, FGR, hydrocephaly), cord prolapse, and **locked twins** (breech first chin locks with cephalic second; 1 in 817).

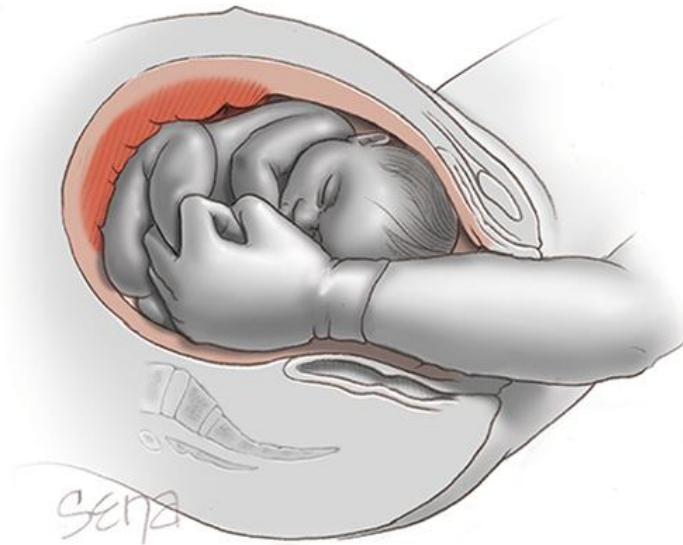


Figure 48-17. Internal podalic version: a hand in the uterus grasps the feet and pulls down while the other hand pushes the head up.

Augmentation and induction

- Active labor and second stage are **slower** with twins.
- Oxytocin augmentation acceptable if criteria are met; induction (oxytocin ± ripening) can be used safely, but ends in **cesarean ~40%** and more maternal morbidity.
- Parkland generally **doesn't induce or augment** twins (overdistended uterus → rupture, PPH). Amniotomy induction is one option.

Vaginal delivery technique

- **Clamp each cord and keep it clamped** until all fetuses are delivered (prevents hypovolemia through anastomoses). Second baby's cord gets **two clamps**; collect cord blood only after all are born.
- Evidence insufficient for or against **delayed cord clamping**.
- After twin 1: assess the second's presentation (abdominal, vaginal, sonographic).
 - **Head/breech fixed in the pelvis** → moderate fundal pressure, **rupture membranes**, recheck for cord prolapse. **Oxytocin if no contractions within ~10 min.**
 - Over the inlet but not fixed → guide it in with vaginal + fundal hands. A shoulder may be converted to cephalic.
 - Not over the inlet or bleeding → **internal podalic version / breech extraction** by a skilled obstetrician with uterine-relaxing anesthesia; otherwise **prompt cesarean**. Avoid delay before the cervix retracts.
- **Intertwin interval classically <30 min**; FHR abnormalities may matter more than the interval itself.

TOLAC

- Uterine rupture and VBAC success similar to singleton TOLAC. ACOG: women with twins and **one prior low transverse cesarean may be TOLAC candidates**. Two thirds deliver both vaginally. **Parkland recommends repeat cesarean.**

Cesarean technique

- **Left lateral tilt** (supine hypotension common).

- **Low transverse hysterotomy** if large enough; Piper forceps for a breech second twin.
- **Low vertical** incision is better if, e.g., a back-down transverse fetus delivers its arms first: easier to extend upward than laterally or as a "T".

Triplets and higher-order

- Monitoring each fetus is hard; later fetuses often need **breech extraction ± internal podalic version**; more cord prolapse and placental separation bleeding.
- Only **17%** attempting vaginal birth delivered vaginally (higher neonatal morbidity). **94%** of US triplets delivered by cesarean.
- **ACOG: vaginal birth may be considered** for uncomplicated triplets if the **first is cephalic**, attendants are experienced and fetuses are **>1500 g**. Parkland delivers most by cesarean.

11. Reducing Fetal Number

Term	Meaning
Multifetal pregnancy reduction (MPR)	Reduce fetal number to improve survival of the rest
Selective reduction	Early removal of a fetus with an anomaly or serious risk
Selective termination	Same indications, but later gestation → greater risk

- Done in the **late 1st trimester**: most spontaneous losses have already happened, fetuses can be assessed, little devitalized tissue remains, low risk of losing the whole pregnancy.
- **Technique**:
 - **Separate chorion** → **intracardiac/intrathoracic KCl**. Don't cross the sacs of retained fetuses.
 - **Monochorionic** → **cord occlusion** (e.g., **RFA** into the intrafetal cord), because of shared vasculature.
- **Counsel on risks**: (1) loss of all fetuses; (2) abortion or retention of the wrong fetus; (3) damage without death; (4) preterm labor; (5) discordance/FGR; (6) maternal complications. Rarely infection, hemorrhage, **DIC** from retained products.

MPR

- **Triplets** → **twins or singleton: less preterm birth <34 wk**, miscarriage not increased. Also benefits higher-order multiples.
- **Twins** → **singleton**: for maternal comorbidity or monochorionic complications.

Selective reduction / termination

- For **discordant anomalies** or severe **TTTS, TAPS, TRAP, sFGR**.
- Usually only when the anomaly is **severe but not lethal**, or when the abnormal fetus threatens the normal one.

Quick-Recall: Chorionicity at a Glance

Feature	Dichorionic diamnionic	Monochorionic diamnionic	Monochorionic monoamnionic
Split timing (if MZ)	<72 h (or any DZ)	4–8 days	8–12 days
Sonography	Twin-peak/lambda , membrane ≥2 mm , 2 placentas or discordant sex	T sign , membrane <2 mm	No membrane , cord entanglement
Anastomoses	0%	~100%	80–90% (AA, protective vs TTTS)
Perinatal mortality	10–20%	30–40%	58–60%
Unique risks	Discordance from implantation/cords	TTTS (10–15%), TAPS (1–6%), TRAP, sFGR, co-twin injury after sIUFD	Cord entanglement , anomalies 18–28%
Growth scans	Every 4 wk from 16–20 wk	Every 2 wk from 16 wk	Every 2 wk; admit 24–28 wk
Weekly testing from	36 wk	32 wk	Daily FHR monitoring inpatient
Deliver at	38⁰/₇–38⁶/₇ wk	34⁰/₇–37⁶/₇ wk	32⁰/₇–34⁰/₇ wk by cesarean

Key Numbers to Memorize

- Twin rate **3.2%**; monozygotic **1 in 250** (constant); twin infant mortality **>4×**, triplets **12×**
- Split **<3 d DCDA**, **4–8 d MCDA**, **8–12 d MCMA**, later conjoined
- Chorionicity accuracy **98%** 1st trimester; dichorionic membrane **≥2 mm**
- Vanishing twin **10–40%**; start multifetal **1 in 8**, born multifetal **1 in 80**
- Blood volume **+50–60%**; CO **+20%** over singletons; weight gain **37–54 lb**; **40–45 kcal/kg/d**
- Aspirin **81 mg** from **12–28 wk**; DVP **<2 cm** oligo, **>8 cm** hydramnios
- TTTS **10–15%** of MC; Quintero **I–V**; scan **q2 wk from 16 wk**; laser **76% vs 51%**; stage ≥III untreated loss **70–100%**
- TAPS: MCA-PSV **>1.5 MoM donor**, **<1.0 MoM recipient**; post-laser TAPS **≤13%**
- TRAP **1 in 35,000**; RFA → **80%** survival
- Discordance **>20%** (ACOG); **>25–30%** predicts harm; fetal death RR **5.6 (>30%)**, **18.9 (>40%)**
- MCDA sIUFD: co-twin death **16×**; neuro morbidity **29%**; deliver if **≥34 wk**
- Preterm: twins **60%**, triplets **98%**; CL **<20 mm** predicts **<34 wk**; PPROM latency **4 vs 7 d**
- Deliver DCDA **38 wk**, MCDA **34–37 wk**, MCMA **32–34 wk**; vaginal breech/noncephalic if **>1500 g and >32 wk**; intertwin **<30 min**
- Triplets: **94%** cesarean; conjoined **1.5/100,000**, thoracopagus most common